

# ALISINA BAYATI

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Citizenship: United States | Iran

## RESEARCH PROFILE

**Research Focus:** Developing **Maximum Entropy Principle (MEP)** frameworks based on deterministic annealing (DA), combined with **control-theoretic tools**, to address NP-hard and nonconvex combinatorial optimization problems with applications to routing, capacitated facility location, constrained sparse linear regression, orthogonal NMF, and safety-constrained dynamic coverage.

**Future Directions:** Bringing control-theoretic perspectives into machine learning by embedding structure, stability, and safety into learning algorithms, including **safety-aware reinforcement learning**, **learning-based control**, and the **control-theoretic analysis and design of optimization and learning algorithms**.

## EDUCATION

### University of Illinois at Urbana–Champaign

Ph.D. Mechanical Science and Engineering

Committee: Prof. Jeff Shamma, Prof. Prashant Mehta, Prof. Yingying Li

2020 – 2026 (Exp.)

Advisor: Prof. Srinivasa Salapaka

### University of Illinois at Urbana–Champaign

M.S. Mathematics

2023 – 2025

GPA: 3.90/4.00

### Sharif University of Technology

B.S. Mechanical Engineering

2015 – 2019

GPA: 3.89/4.00 (18.10/20), Advisor: Prof. Hamed Moradi

## PUBLICATIONS

### Peer-Reviewed

1. **A. Bayati**<sup>†</sup>, D. Tiwary<sup>†</sup>, S. Salapaka. “A Control Barrier Function Approach to Constrained Resource Allocation Problems in a Maximum Entropy Principle Framework.” *IEEE Conference on Decision and Control (CDC)*, 2025. 📄 [Paper](#) 🔄 [Code](#)
2. S. Basiri, **A. Bayati**, S. Salapaka. “Orthogonal Non-negative Matrix Factorization with Sparsity Constraints.” *IEEE Conference on Decision and Control (CDC)*, 2025. 📄 [Paper](#) 🔄 [Code](#)
3. **A. Bayati**, A. Srivastava, V. Mundada, S. Salapaka, H. Feng, A. Malvandi. “Energy Efficiency in Industrial Drying: A Hybrid Ultrasonic System with a Novel Dynamic Optimization Framework.” *Energy Conversion & Management*, 2025. 📄 [Paper](#)
4. A. Srivastava<sup>†</sup>, **A. Bayati**<sup>†</sup>. “Sparse Linear Regression with Constraints: A Flexible Entropy-Based Framework.” *European Control Conference (ECC)*, 2024. 📄 [Paper](#)
5. **A. Bayati**, A. Srivastava, A. Malvandi, H. Feng, S. Salapaka. “Towards Efficient Modularity in Industrial Drying: A Combinatorial Optimization Viewpoint.” *American Control Conference (ACC)*, 2023. 📄 [Paper](#)

### Working Papers

6. **A. Bayati**, A. Srivastava, S. Salapaka. “Adaptive Resource Allocation under Demand Distribution Shifts: A Maximum Entropy Framework with Safety and Mobility Constraints.”

## RESEARCH EXPERIENCE

### Graduate Research Assistant, SENSIC Lab, University of Illinois Urbana–Champaign

2020 – Present

#### • Control-Theoretic Approaches to Combinatorial Optimization

2024 – Present

- Designed a dynamic feedback strategy for constrained optimization that provably drives decision variables toward KKT stationarity with guarantees on convergence and feasibility.
- Applied to capacitated facility location problems, achieving up to  $\sim 4\times$  and  $\sim 74\times$  speedups over Safe Gradient Flow (Cortés et al.) and SciPy’s SLSQP. 📄 [Paper](#) 🔄 [Code](#)
- Applied the framework to orthogonal non-negative matrix factorization with feature sparsity. 📄 [Paper](#) 🔄 [Code](#)
- Proved almost-sure avoidance of strict saddle points under random initialization.
- Ongoing work extends these frameworks to non-stationary environments with shifting demand distributions, aiming to track time-varying optimizers with high sample efficiency.

#### • Mixed-Integer Control Frameworks for Hybrid Actuator Systems

2020 – Present

- Developed a mixed-integer multi-stage optimization framework for ultrasonic/convective drying, reducing energy consumption by **13%** relative to a static single-stage baseline while satisfying product quality constraints (moisture and color). 📄 [Journal Paper](#) 📄 [ACC Paper](#)
- Extending the framework to smart drying systems involving multiple drying technologies, jointly optimizing both the drying sequence and operating conditions.
- Integrating mesh-based graph neural networks trained offline to replace CFD/statistical models for fast differentiable inference.

- **Routing Optimization in Residential Wi-Fi Mesh Networks** 2021 – 2023
  - Built a real-time digital twin for Wi-Fi mesh networks using permutation-invariant neural networks to solve large-scale placement and routing problems.
  - Achieved network latency in the **lowest 3rd percentile** across 50,000 uniformly sampled brute-force configurations.
  - Project funded by *Foxconn Interconnect Technology (FIT)*.

#### Undergraduate Researcher, Sharif University of Technology

2018 – 2019

- Designed optimal PID controllers for nonlinear cancer drug-delivery models in MATLAB/Simulink to minimize side effects while achieving target tumor reduction.

## TEACHING & MENTORSHIP

### Graduate Teaching Assistant, University of Illinois Urbana–Champaign

2023 – Present

- **Recognition:** Ranked “Excellent” in campus-wide student evaluations (2024).
- **Graduate Course:** *TAM 541 – Mathematical Methods I*. Led recitation sessions for 50+ graduate students covering complex analysis, linear algebra, and ordinary differential equations.
- **Undergraduate Courses:** *ME 340 – Dynamics of Mechanical Systems*, *TAM 210/211 – Statics & Intro to Dynamics*. Managed laboratory sections and incorporated MATLAB-based simulations to support conceptual understanding.

### Research Mentorship, SENSIC Lab (UIUC)

2024 – Present

- Mentoring two undergraduate researchers on routing optimization for residential Wi-Fi mesh networks.

## SELECTED GRADUATE COURSEWORK

| Mathematics                          | Machine Learning & Statistics      | Control & Systems             |
|--------------------------------------|------------------------------------|-------------------------------|
| Real Analysis                        | Statistical Reinforcement Learning | Theory of Optimal Control     |
| Functional Analysis                  | Machine Learning                   | Analysis of Nonlinear Systems |
| Abstract Algebra                     | Computer Vision                    | Linear Control Systems        |
| Abstract Linear Algebra              | Probability & Statistics           | Dynamic Programming           |
| Optimization by Vector-Space Methods |                                    | Random Processes              |

## HONORS & AWARDS

- **Best Presentation Award**, Optimization & Reinforcement Learning Session, CSL Student Conference 2026
- **IEEE Student Travel Award**, Conference on Decision and Control (CDC) 2025
- **IEEE Student Travel Award**, American Control Conference (ACC) 2023
- **Teachers Ranked as Excellent**, campus-wide student evaluations, University of Illinois Urbana–Champaign 2024
- **Graduate Admission Fellowship (First Year)**, Mechanical Science & Engineering Department, UIUC 2020
- **Top 0.2% National Rank**, Iran Nationwide University Entrance Exam (182,000+ participants) 2015

## TECHNICAL SKILLS

|                         |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| <b>Programming</b>      | Python (NumPy, SciPy, pandas, PyTorch, CVXPY, scikit-learn), MATLAB, C, $\LaTeX$ |
| <b>Scientific Tools</b> | Jupyter, Simulink, OpenCV                                                        |
| <b>Languages</b>        | Persian (Native), English (Fluent), Arabic (Basic)                               |

## PROFESSIONAL SERVICE

**Reviewer:** IEEE Conference on Decision and Control (CDC) 2025; American Control Conference (ACC) 2024; European Control Conference (ECC) 2024; Learning for Dynamics and Control (L4DC) 2025.

## REFERENCES

Available upon request.